UNIT 3 – LINKED LIST

TOPIC: CANTEEN ORDER PROCESSING USING LINKED LIST

1. Introduction

A linked list is a linear data structure consisting of nodes connected using pointers. Each node stores data and a link to another node.

This project implements a Canteen Order Processing system using three types of linked lists: Singly Linked List, Doubly Linked List, and Circular Linked List. It is developed using C++ and allows users to insert, display, search, and delete food orders.

2. Problem Statement

Develop a Canteen Order Processing System using a menu-driven C++ program that implements Singly Linked List, Doubly Linked List, and Circular Linked List to manage student food orders. The system should perform insertion, deletion, searching, and traversal operations to efficiently handle canteen orders.

3. Objectives

1. To implement SLL, DLL, and CLL in C++.
2. To perform insertion at the beginning, end, and a specified position.
3. To perform deletion by Order ID and specified position.
4. To search for specific food orders.
5. To perform different traversal operations.
6. To understand dynamic memory allocation.

4. Data Structures Used

4.1 Singly Linked List (SLL)

Each node contains order details and a pointer to the next node. Traversal is possible in the forward direction.

4.2 Doubly Linked List (DLL)

Each node contains order details and two pointers: one to the previous node and another to the next node. Traversal is possible in both directions.

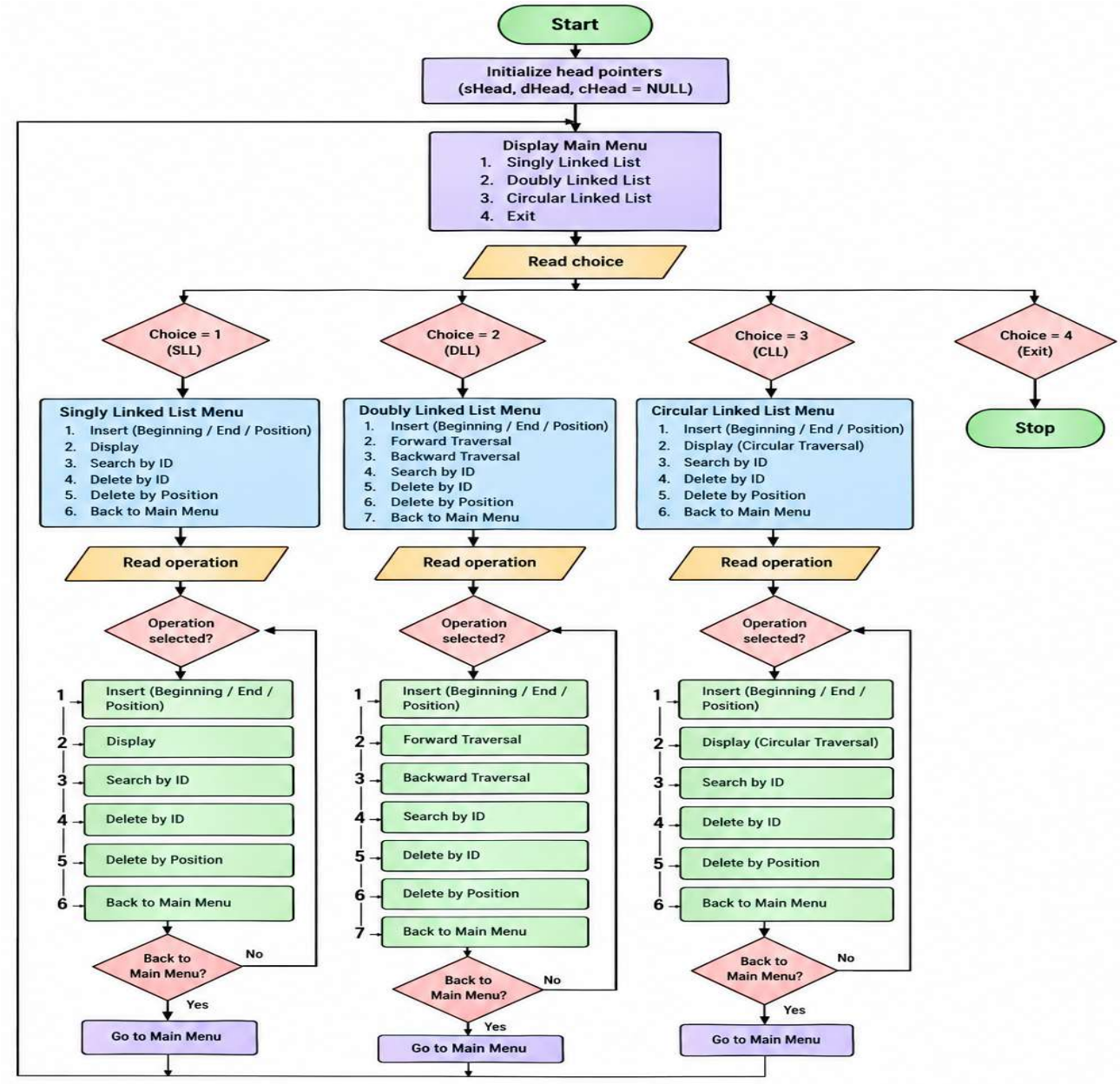
4.3 Circular Linked List (CLL)

In a circular linked list, the last node points back to the first node. Traversal continues until the starting node is reached again.

5. Algorithm

1. Start the program.
2. Initialize the head pointers.
3. Display the main menu.
4. Select SLL, DLL, CLL, or Exit.
5. Display the selected linked list's menu.
6. Choose an operation.
7. Perform insertion, deletion, searching, or traversal.
8. Display the result.
9. Repeat the operations until returning to the main menu.
10. Repeat the main menu until Exit is selected.
11. Stop the program.

6. Flowchart



7. Working of the Project

The program displays a menu to select one of the three linked lists.

- SLL: Supports insertion, display, searching, and deletion.
- DLL: Supports insertion, forward traversal, backward traversal, searching, and deletion.
- CLL: Supports insertion, circular traversal, searching, and deletion.

Insertion can be performed at the beginning, end, or a specified position. Orders can be deleted using the Order ID or a specified position.

Each order stores an Order ID, Student Name, and Food Item. The program uses dynamic memory allocation to create nodes when required.

8. Pogramming Language and Concepts

Concepts used:

- Programming Language: C++
- Structures
- Pointers
- Dynamic Memory Allocation
- Singly Linked List
- Doubly Linked List
- Circular Linked List
- Insertion and Deletion
- Searching and Traversal

9. Applications

- Managing food orders in a college canteen.
- Maintaining student order records.
- Practising dynamic data structures.
- Understanding linked list operations in real-life applications.

10. Advantages

1. Dynamic memory allocation.
2. Easy insertion and deletion of nodes.
3. Different traversal methods.
4. Practical understanding of linked lists.
5. Menu-driven and user-friendly program.

11. Conclusion

Thus, we successfully implemented a Canteen Order Processing program in C++ using Singly Linked List, Doubly Linked List, and Circular Linked List. The project demonstrates insertion, deletion, searching, and traversal operations and improves our understanding of linked lists and dynamic memory allocation.